simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=749

SIMATS ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

AL NAJEEB A
192424325RB

CO2-AT1-Scenario Based Program

Back

QUESTIONS

- Q1 Student Registration System 10 marks
- Q2 E-Commerce Cart System 10 marks
- Q3 Library Management System 10 marks
- Q4 Employee Data Processing 10 marks
- Q5 Online Voting System 10 marks

5/5 answered

Q5 Online Voting System

10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.
(a) Identify three suitable Collection types for this system.
(b) Design a structure using Set and Map for voters and vote counting.
(c) Write a Java program using HashSet and HashMap to:
• Record votes
• Prevent duplicate voting • Display results

Your Code

```java
import java.util.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        HashSet<String> voted = new HashSet<>();
        HashMap<String, Integer> count = new HashMap<>();

        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 0; i < n; i++) {
            String voter = sc.nextLine();
            if (voted.contains(voter)) {
                System.out.println("Duplicate vote");
            } else {
                voted.add(voter);
                String cand = sc.nextLine();
                if (count.containsKey(cand)) {
                    int c = count.get(cand);
                    count.put(cand, c + 1);
                } else {
                    count.put(cand, 1);
                }
            }
        }
        System.out.println("Results");
        for (String key : count.keySet()) {
            System.out.println(key + " : " + count.get(key));
        }
    }
}
```

Input

```
1
ram
jhon
```

Output

Run Save

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Q2 E-Commerce Cart System

10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.
(a) List three Collection classes applicable for cart management.
(b) Represent how cart items are stored using a List-based structure.
(c) Write a Java program using ArrayList and Iterator to:
• Add items
• Remove an item
• Display all items

Your Code

```java
import java.util.*;
public class CartDemo {
  public static void main(String[] args) {
    ArrayList<String> cart = new ArrayList<>();
    cart.add("Pen");
    cart.add("Book");
    cart.add("Bag");

    System.out.println("Before removal:");
    Iterator<String> it1 = cart.iterator();
    while (it1.hasNext()) System.out.println(it1.next());

    String item = "Book";
    int pos = -1;
    for (int i = 0; i < cart.size(); i++) {
      if (cart.get(i).equalsIgnoreCase(item)) {
          pos = i;
          break; }
    }

    if (pos != -1) {
      for (int i = pos; i < cart.size() - 1; i++) cart.set(i, cart.get
      cart.set(cart.size() - 1, "");
    }

    System.out.println("After removal:");
    Iterator<String> it2 = cart.iterator();
    while (it2.hasNext()) {
        String s = it2.next();
        if (!s.equals("")) System.out.println(s);
    }
  }
```

Input

stdin input

Output

```
Before removal:
Pen
Book
Bag
After removal:
Pen
```

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5/5 answered

Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.
(a) Name three Map implementations in Java.
(b) Design a schema-like representation using Map for storing ISBN and book details.
(c) Write a Java program using HashMap to:
• Insert book details
• Retrieve a book
• Display all books

Your Code

```java
import java.util.*;
public class Main {
  public static void main(String[] args) {
    HashMap<String, String> library = new HashMap<String, String>();
    library.put("111", "Java Basics - najeeb");
    library.put("222", "DSA Using Java - Alnajeeb");
    library.put("333", "Computer Networks - najee");
    String isbn = "222";
    System.out.println("Retrieve ISBN: " + isbn);
    if (library.containsKey(isbn)) {
      System.out.println("Book: " + library.get(isbn));
    } else {
      System.out.println("Book not found");
    }
    System.out.println("All books:");
    for (String key : library.keySet()) {
      System.out.println("ISBN: " + key + " -> " + library.get(key));
    }
  }
}
```

Input

stdin input

Output

```
Retrieve ISBN: 222
Book: DSA Using Java - Alnajeeb
All books:
ISBN: 111 -> Java Basics -
najeeb
ISBN: 222 -> DSA Using Java -
```

Run Save

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5/5 answered

Q4 Employee Data Processing

10 marks

Suppose that a company processes employee records and filters employees based on salary.
(a) List three advantages of Generics in Collections.
(b) Represent how employee data is stored using List with Generics.
(c) Write a Java program using Iterator to:
• Store employee objects
• Display employees with salary > 50,000

Your Code

```java
import java.util.*;
public class Main {
    public static void main(String[] args) {
        List<Employee> list = new ArrayList<Employee>();
        list.add(new Employee(1, "naj", 45000));
        list.add(new Employee(2, "najee", 60000));
        list.add(new Employee(3, "naje", 52000));
        list.add(new Employee(4, "najeeb", 30000));

        Iterator<Employee> it = list.iterator();

        while (it.hasNext()) {
            Employee e = it.next();
            if (e.salary > 50000) {
                System.out.println(e.id + " " + e.name + " " + e.salary
            }
        }
    }
}
class Employee {
    int id;
    String name;
    double salary;

    Employee(int id, String name, double salary) {
        this.id = id;
        this.name = name;
        this.salary = salary;
    }
}
```

Input

stdin input

Output

```
2 najee 60000.0
3 naje 52000.0
```

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        for (int i = 0; i < n; i++) {
            String voter = sc.nextLine();
            if (voted.contains(voter)) {
                System.out.println("Duplicate vote");
            } else {
                voted.add(voter);
                String cand = sc.nextLine();
                if (count.containsKey(cand)) {
                    int c = count.get(cand);
                    count.put(cand, c + 1);
                } else {
                    count.put(cand, 1);
                }
            }
        }
        System.out.println("Results");
        for (String key : count.keySet()) {
            System.out.println(key + " : " + count.get(key));
        }
    }
}
```

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